

that the recursive algorithm uses. The following *exponentWithPowerRule.py* program has such a function:

Python

```
def exponentWithPowerRule(a, n):
    # Step 1: Determine the operations to be performed.
    opStack = []
    while n > 1:
        if n % 2 == 0:
            # n is even.
            opStack.append('square')
            n = n // 2
        elif n % 2 == 1:
            # n is odd.
            n -= 1
            opStack.append('multiply')

    # Step 2: Perform the operations in reverse order.
    result = a # Start result at `a`.
    while opStack:
        op = opStack.pop()

        if op == 'multiply':
            result *= a
        elif op == 'square':
            result *= result

    return result

print(exponentWithPowerRule(3, 6))
print(exponentWithPowerRule(10, 3))
print(exponentWithPowerRule(17, 10))
```

Here is the equivalent JavaScript program in *exponentWithPowerRule.html*:

JavaScript

```
<script type="text/javascript">
function exponentWithPowerRule(a, n) {
    // Step 1: Determine the operations to be performed.
    let opStack = [];
    while (n > 1) {
```